

Intelligent Sales Forecasting & Demand Intelligence

Executive Summary

Over the past four years, sales have grown steadily and now show a repeatable, predictable seasonal pattern: demand consistently peaks in September and again in November-December, and dips every January-February. Using this pattern, we built and compared three forecasting approaches and found one — a statistical method called SARIMA — consistently the most accurate, with an average forecast error of about 21% on recent months. We also built an early-warning system that automatically flags unusual sales weeks, and a system that groups our 17 product sub-categories into four demand patterns, each with its own recommended stocking approach. All of this is available in a live, interactive dashboard that updates automatically and requires no coding knowledge to use.

Key Findings

- **Growth with strong seasonality:** Total revenue grew from roughly \$484K (2015) to \$733K (2018). Technology is the top revenue category (~\$836K total), narrowly ahead of Furniture and Office Supplies.
- **Predictable seasonal swings:** September, November, and December are reliably the strongest months every year; January and February are reliably the weakest. This pattern held in all four years of data.
- **Shipping is consistent:** Orders ship in about 4 days on average, with less than a 4-hour difference between the fastest and slowest region — shipping speed is not a regional problem worth prioritizing right now.
- **The West region leads in total sales**, but the East region is forecasted to grow fastest going into next quarter (see Segment Forecasts below).

3-Month Sales Forecast

We tested three forecasting methods head-to-head on real historical data. The table below shows how far off each model's predictions were from actual results, in dollars (MAE, RMSE) and as a percentage (MAPE) — lower is better.

Model	MAE	RMSE	MAPE	Recommended?
SARIMA	\$19,372	\$19,947	20.7%	Yes — production model
Prophet	\$21,927	\$22,725	24.7%	Backup / stakeholder-facing
XGBoost	\$51,137	\$59,374	51.1%	Not yet — needs more data

In plain terms: SARIMA's forecasts were off by about \$19,000 per month on average during our test period, versus roughly \$22,000 for Prophet and \$51,000 for the machine-learning model. We recommend SARIMA as the production forecasting model going forward.

Segment Forecasts — Where Growth Is Coming From

Applying the winning model to individual categories and regions, and comparing the forecast to the same three months one year earlier, shows where growth is strongest:

Segment	YoY Growth (forecast)
East (Region)	+157.8%
Furniture (Category)	+49.6%
Office Supplies (Category)	+34.5%
Technology (Category)	+26.6%
West (Region)	+12.3%

Top Anomalies Detected

An automated early-warning system flags weeks where sales were unusually high or low. Three notable spikes:

- Late November 2018 and December 2018 — the two largest anomalies detected, consistent with Black Friday/Cyber Monday and holiday shopping.
- Mid-March 2015 — an early, unusually large spike, most likely a handful of large bulk orders (e.g., office copiers/machines) rather than a broad demand shift; worth a manual check for accuracy.
- Two independent detection methods agreed on only a small handful of weeks — the weeks flagged by both are the highest-confidence anomalies and the best use of a manual review.

Product Demand Segments & Stocking Strategy

Products were grouped into four demand patterns based on sales volume, growth, and volatility:

Cluster	Example Sub-Categories	Stocking Strategy
High Volume, Stable Demand	Chairs, Phones, Storage, Binders, Tables, Accessories	Steady safety stock, automatic reorder
Growing Demand	Copiers, Appliances, Bookcases, Paper, and 8 others	Increase stock ahead of forecast; watch lead times
Declining Demand	Machines	Scale back orders; consider clearance

Business Recommendations

- **1. Adopt SARIMA as the standing monthly forecast**, refreshed each month as new sales data comes in, to guide inventory purchasing decisions 1-3 months ahead.
- **2. Pre-position inventory ahead of the Sept/Nov/Dec peak** and deliberately draw down stock in Jan/Feb — this pattern has held for four consecutive years and is low-risk to plan around.
- **3. Prioritize the East region and Furniture category** for the next growth push, and review Machines for reduced future ordering given its declining-demand signal.

Risk & Limitation

This system is built on four years of historical data and works best for short-term forecasts (1-3 months). It has not been tested through a major external shock (e.g., a supply chain disruption or a new competitor), and the machine-learning model in particular needs more historical data before it can be trusted for production use. We recommend treating all forecasts as a planning aid alongside — not a replacement for — manager judgment, and re-validating model accuracy every quarter as new data arrives.